

Deepak Sathyan

arXiv | InspireHEP | Google Scholar | ORCID | Website

✉ dsathyan@tamu.edu

📍 Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

Education

University of Maryland

Ph.D. in Physics

Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass

Advisor: Dr. Kaustubh Agashe

College Park, MD

Aug 2018 – July 2024

Boston University

Bachelor of Arts in Physics, Summa Cum Laude with Honors

GPA: 3.9

Minor: Mathematics

Thesis: Muon ($g-2$): Searching for the Muon Electric Dipole Moment

Advisor: Dr. Robert Carey

Boston, MA

Aug 2014 – May 2018

Experience

Postdoctoral Research Associate

Mitchell Institute for Fundamental Physics and Astronomy
Texas A&M University

College Station, TX

Sept 2024 – Aug 2027

3 years

Research Interests

Beyond the Standard Model searches with Collider Physics

searches at the LHC, sub-GeV dark matter, invisible new physics via precision measurements, Kaluza-Klein modes in Randall-Sundrum models, baryon number violation models, muon collider phenomenology

Searches for sub-GeV Dark Matter

Constraints on dark matter-neutrino interactions, dark photon production at the LHC, dark sector production at muon collider, astrophysical signals of cosmic ray-dark matter scattering, astrophysical sources of sub-GeV dark matter

Neutrino Physics

Supernova neutrino attenuation and detection, neutrino oscillations in open systems, neutrinophilic mediators

Articles in Review

3. Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering

B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan

arxiv.org/abs/2605.08010

May 2026

- | | |
|--|------------------|
| <p>2. A Baryon and Lepton Number Violation Model Testable at the LHC
 A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan
 arxiv.org/abs/2508.21064</p> | <p>Aug 2025</p> |
| <p>1. New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis
 P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
 arxiv.org/abs/2507.01000</p> | <p>July 2025</p> |

Articles in Refereed Journals

- | | |
|--|------------------|
| <p>5. New laboratory constraints on neutrinophilic mediators
 P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
 arxiv.org/abs/2407.12738 (Phys.Lett.B 868 (2025) 139765)</p> | <p>July 2024</p> |
| <p>4. ‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m_W
 K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
 arxiv.org/abs/2404.17574 (JHEP 02 (2025) 139)</p> | <p>Apr 2024</p> |
| <p>3. A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET
 K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
 arxiv.org/abs/2310.13687 (Phys.Lett.B 855 (2024) 138774)</p> | <p>Oct 2023</p> |
| <p>2. Energy-peak based method to measure top quark mass via B-hadron decay lengths
 K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan
 arxiv.org/abs/2212.03929 (JHEP 06 (2023) 021)</p> | <p>Dec 2022</p> |
| <p>1. LHC Signals for KK Graviton from an Extended Warped Extra Dimension
 K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan
 arxiv.org/abs/2008.06480 (JHEP 11 (2020) 109)</p> | <p>Aug 2020</p> |

Articles in Preparation

- | | |
|---|------------------|
| <p>1. LHC as a Beam Dump: Probing Light Dark Photons
 B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan
 arxiv.org/a/sathyan_d_1</p> | <p>Sept 2026</p> |
| <p>2. New Probes of Dark Sectors at a Muon Collider
 B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan
 arxiv.org/a/sathyan_d_1</p> | <p>Jan 2027</p> |
| <p>3. Exploring open systems approach to neutrino oscillations
 A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou
 arxiv.org/a/sathyan_d_1</p> | <p>Mar 2027</p> |

4. **LHC Signals for KK Gravitons Producing 8-Particle Final States** Mar 2027
K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan
arxiv.org/a/sathyan_d_1

Ph.D. Thesis

1. **Unifying Searches for New Physics with Precision Measurements of the W Boson Mass** 2024
D. Sathyan
drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)

Snowmass2021 Contributions

5. **TF07 Snowmass Report: Theory of Collider Phenomena** Oct 2022
F. Maltoni et al.
arxiv.org/abs/2210.02591
4. **Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021** Sept 2022
K. Agashe et al.
arxiv.org/abs/2209.11267
3. **Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021** Sept 2022
T. Bose et al.
arxiv.org/abs/2209.13128
2. **Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond** Apr 2022
K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan
arxiv.org/abs/2204.02928
1. **Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models** Mar 2022
K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan
arxiv.org/abs/2203.13305

Muon g-2 Articles

3. **The straw tracking detector for the Fermilab Muon g-2 Experiment** Nov 2021
B.T. King et al.
arxiv.org/abs/2111.02076 (JINST 17 (2022) P02035)
2. **Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab** Apr 2021
Muon g-2 Collaboration
arxiv.org/abs/2104.03240 (Phys.Rev.Accel.Beams 24 (2021) 044002)

1. **Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm**
Muon g-2 Collaboration
arxiv.org/abs/2104.03281 (Phys.Rev.Lett. 126 (2021) 141801)

Apr 2021

Seminars

7. **CMS Exotica general meeting, May 2024**
Unification of Searches and Measurements: Probing BSM with the W boson mass measurement
6. **Seminar at Texas A&M, December 2023**
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
5. **Seminar at UT Austin, December 2023**
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
4. **CMS SUSY general meeting, November 2023**
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
3. **HEP Seminar at University of Pittsburgh, November 2023**
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
2. **HEP Seminar at Northwestern University, October 2023**
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
1. **Theory Seminar at Washington University in St. Louis, October 2023**
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

Conference Talks

14. **Light Dark World 2026 at Carleton University, July 2026**
Can the LHC be sensitive to the Light Dark World?
13. **CETUP* Workshop, June 2026**
Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering
12. **Particle Physics on the Plains at the University of Kansas, November 2025**
A Baryon and Lepton Number Violation Model Testable at the LHC
11. **CETUP* Workshop, June 2025**
Can the LHC be sensitive to light dark mediators?
10. **Phenomenology Conference at the University of Pittsburgh, May 2025**
Can the LHC be sensitive to light dark mediators?

9. **Particle Physics on the Plains at the University of Kansas, November 2024**
A comprehensive analysis of supernova neutrino-dark matter interactions
8. **Texas TACOS @ UT Austin, October 2024**
A comprehensive analysis of supernova neutrino-dark matter interactions
7. **NuFact Conference, September 2024**
A comprehensive analysis of supernova neutrino-dark matter interactions
6. **Mitchell Conference at Texas A&M University, May 2024**
A comprehensive analysis of supernova neutrino-dark matter interactions
5. **Particle Physics on the Plains at the University of Kansas, October 2023**
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
4. **Phenomenology Conference at the University of Pittsburgh, May 2023**
Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos
3. **Phenomenology Conference at the University of Pittsburgh, May 2022**
Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I)
2. **April APS Meeting, April 2021**
Signals of KK graviton from extended warped extra dimensions at the LHC (II)
1. **Phenomenology Conference at the University of Pittsburgh, May 2020**
Signals of KK graviton from extended warped extra dimensions at the LHC (II)

Conferences Organized

- The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026**
Texas A&M University, May 27-30, 2026
- The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025**
Texas A&M University, May 13-16, 2025

Awards

- | | |
|--|----------|
| Breakthrough Prize in Fundamental Physics | Apr 2026 |
| Muon g-2 Collaboration | |

Schools Attended

- | | |
|---|-----------------------|
| GGI Lectures on the Theory of Fundamental Interactions | Jan 2023 – Jan 2023 |
| TASI 2024: The Frontiers of Particle Theory | June 2024 – June 2024 |

Outreach

BU Student Mentorship Program PRISM (2016-2018)

Mentored incoming freshman undergraduates in physics, providing guidance on courses, research opportunities, and resources

TAMU Physics and Engineering Festival (2025-2027)

Presented demonstrations of fluid dynamics to the general public, including high school students and families

University of South Dakota talk: "My path through particle physics research so far" (December 2024)

Presented my path through particle physics research to students, showing a variety of interesting ideas to pursue

Skills

Programming:

Mathematica, Python, C++

Physics Software:

MadGraph5_aMC@NLO, Pythia6, Pythia8, Delphes, ROOT, GEANT4, FeynRules